

Mask building for perceptually hiding frequency embedded watermarks

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Abstract

The interest for image watermarking techniques has rapidly grown during recent years. Two requirements are needed to be satisfied to use watermarking techniques for copyright protection: unperceivability and robustness against image processing algorithms and forgery attacks. In particular, it is widely accepted that the exploitation of the characteristics of the Human Visual System should greatly help for satisfying both these requirements. In this paper some solutions to the problem of building some perceptual masks for better hiding watermarks embedded in the full-frame DCT domain are presented. The results support the validity of the approach.

1 Introduction

Efficient techniques for image watermarking have to accurately balance between two contrasting requirements: on one side, the hidden watermark should be invisible, i.e., unperceivable to the Human Visual System (HVS), on the other side, it should not be inserted in the areas of the image which are few perceptually important, given that these would be discharged (and the watermark with them) by those image compression and image enhancement techniques designed for taking into account the characteristics of the HVS (e.g. JPEG, median and despeckle filtering). The exploitation of the characteristics of the HVS is, thus, a very important and delicate task to be attacked for implementing effective image watermarking tools [1].

In this paper the problem of perceptually masking the watermarks embedded in the full frame Discrete Cosine Transform (DCT) [2, 3] is addressed. In particular some possible masking images, suitable for reducing watermark strength over those regions of the image that are more sensitive to disturbs, are presented.

The paper is organized as it follows: in Section 2 the main characteristics of the HVS are reviewed and the problems encountered while trying to exploit them for full-frame-DCT-based watermarking are presented; in Section 3 a solution for the just mentioned problems

is presented based on the use of spatial insensitivity masks; in Section 4 some possible approaches for building such masks are proposed; finally, in Section 5 experimental results are presented and discussed, which support the validity of the methodology.

2 Main characteristics of the HVS

The HVS is an imperfect device: a visual signal, for example, must have, to be perceived, a minimum amount of contrast which depends on its mean luminance and on its frequency. Furthermore an informative signal of a given frequency can hide a disturbing signal of similar frequencies; in particular the contrast visibility threshold of a sinusoidal signal superimposed to an iso-frequency grating, is approximately proportional to the contrast of the grating itself [4] (*iso-frequency masking*). Such characteristic is implicitly exploited by those techniques [2, 3] embedding the watermark in the full frame DCT domain. In [3], for example, the DCT coefficients belonging to a fixed subset of the full-frame DCT of an image I , are modified, for carrying the watermark, according to the rule:

$$y_j = x_j + \alpha w_{ij} |x_j| \quad (1)$$

where $\{x_j\}_{j=0,L-1}$ are the DCT coefficients to be modified, $\{w_{ij}\}_{j=0,L-1}$ is the i^{th} watermark and α is a parameter accounting for the strength of the watermark. The coefficients y_j are re-inserted in place of the x_j , and, by inverse DCT, the watermarked image I' is obtained. In this way, to each sinusoid which is present in the image (masking signal), another sinusoid (disturb/watermark) is added, having amplitude proportional to that of the masking signal itself. Given that α is properly tuned, masking in the frequency domain is, thus, achieved.

However such an approach lacks in spatial localization; in fact, the disturbing signal, inserted in the DCT domain, is spread over the whole image, also where the masking signal is not present: in these areas (e.g. uniform areas) masking is ineffective. On the contrary, data hiding techniques based on block

DCT [5, 6], can perform locally adaptive masking just because of their block-wise architecture (masking is performed block by block). Nevertheless some good characteristics of techniques based on full-frame DCT (e.g. their intrinsic robustness against image resizing [7]) motivate to investigate how some masking localization can be obtained also for them.

3 Spatial masking

The approach we propose is to construct another watermarked image I'' by mixing the original image I and its watermarked version I' :

$$I'' = (1 - M) I + M I' \quad (2)$$

where the masking image M has values in the interval $[0, 1]$ and gives a measure (for each point of the original image I) of its insensitivity to noise. Given that $W = I' - I$ is the watermark image, equation (2) can be rewritten as

$$I'' = I + MW \quad (3)$$

i.e. the watermark image is multiplied by the masking image. From the theory of Fourier transform, this corresponds to perform (in the DCT domain) a sort of convolution between the DCT of the masking image and the watermarking signal $\{w_{ij}\}_{j=0,L-1}$; thus given that, for recovering the watermark, its autocorrelation characteristics are exploited [3, 7], watermark detection should be more difficult. In particular, this suggests that the spectral content of the masking images should be as much concentrated as possible to low frequencies, in order not to degrade too much the watermark (ideally the DCT of the masking image should be a delta function). On the other side, experimental results showed that, by using such masking approach, it is possible to embed, into the images, watermarks of higher energy, than it was possible without masking, by keeping them equally unperceivable. Thus, also watermark detection benefits from masking. Furthermore, if the masking image is properly designed, the watermark energy is concentrated (into images) in those areas (edges, high activity regions) which are less affected by those algorithms (JPEG compression, median and despeckle filtering) that take into account the characteristics of the HVS: a good gain in robustness is, thus, also achieved.

Watermark detection is based on the computation of the correlation between the DCT coefficients of the watermarked (and possibly corrupted) image I'' , and the watermarking code [3, 7]. The value of the correlation is compared with a threshold which can be set based on some statistical parameters of the marked

image, in such a way to grant a given probability of false positive detection [8].

Some considerations are needed regarding the amount of energy that is inserted in the image. By observing equation (3) it is evident that, by masking the watermark, its energy depends also on the masking image. In particular a mean value of α , given by

$$\bar{\alpha} = \alpha \bar{M} \quad (4)$$

where

$$\bar{M} = \frac{1}{N_x N_y} \sum_{u=0}^{N_y-1} \sum_{v=0}^{N_x-1} M(u, v), \quad (5)$$

has to be considered. It will be later shown that larger values of $\bar{\alpha}$ can be tolerated, without deteriorating image quality, while using the masking techniques.

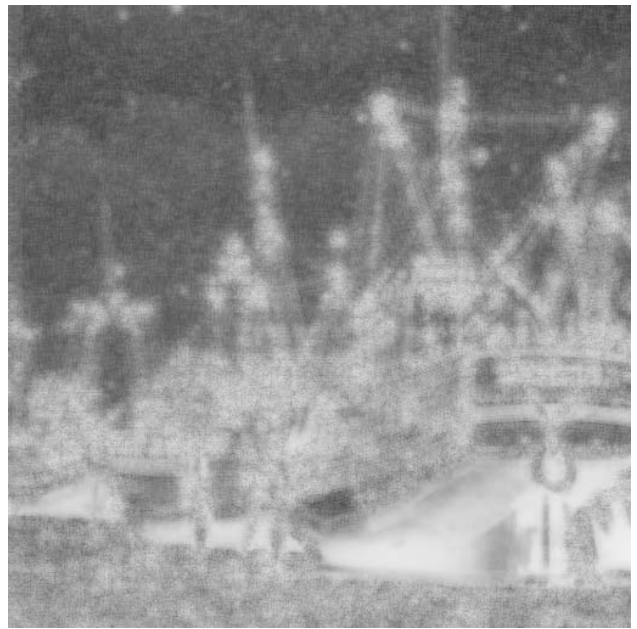


Figure 1: Masking image M_s built based on the theoretical model of the HVS.

4 Building masking images

Some possible forms of the masking image are proposed in the following: some are built based on heuristic considerations, others are, on the contrary, obtained by using a multiple channel model of the HVS similar to that proposed in [9].

Based on a multiple channel model of the HVS [9], a theoretically motivated sensitivity map (M_s) can be estimated (see Figure 1). This model assumes that the human eye partitions the frequency domain in a set of

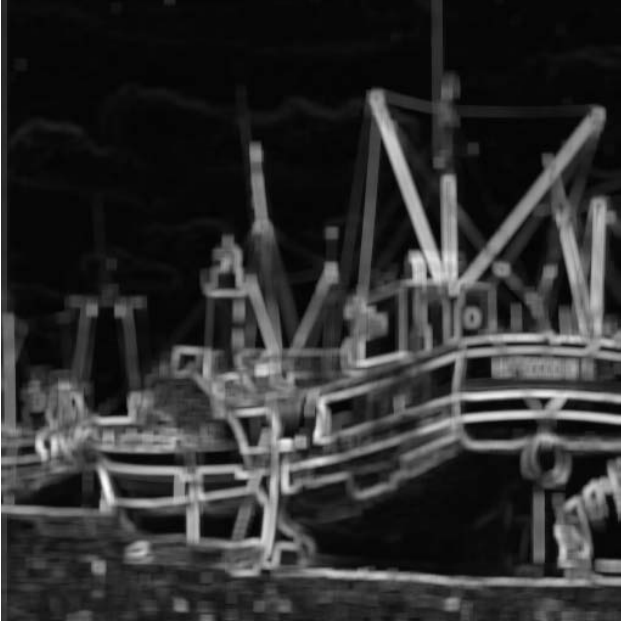


Figure 2: Masking image M_σ built based on the computation of the local variance.

non-overlapping subbands, and that the image component in each subband only mask disturbing signals in that subband. In particular, in order to mask the watermark, such a sensitivity map has been computed by considering only those image subbands that covers the set of frequencies in the DCT domain where the watermark is inserted.

For the derivation of the heuristic masks, some assumptions are made [10]: first of all the regions of high activity are highly insensitive to disturbs; second the edges (objects contours) are more sensitive to distortions than the highly textured areas; finally it is considered that darker and brighter regions of the image are less sensitive to noise.

Based on these hypotheses, a first simple mask (M_σ) can be built by computing, at each image location, the local variance over a square window W (e.g. 9×9), and by normalizing the obtained image with respect to its maximum value. The value of the mask at each pixel is thus:

$$M_\sigma(x, y) = \frac{1}{M_{max}} \sum_{(i,j) \in W} [I(i, j) - \mu_W(i, j)]^2 \quad (6)$$

where $\mu_W(i, j)$ is the local mean computed in the window W centered at the pixel position (x, y) , and $M_{max} = \max_{m,n} M_\sigma(m, n)$ is the maximum value of the local variance over the whole image. The main weakness of this mask is that it does not differentiate

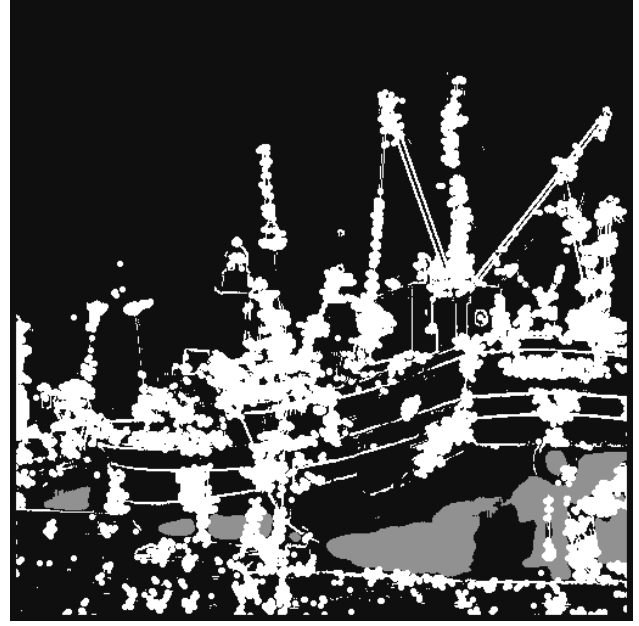


Figure 3: Masking image M_{bp} built by the system sketched in Figure 4.

between highly textured regions and image edges (on the contrary the latter will produce, often, a higher value of local variance) as can be seen in Figure 2. Furthermore this mask do not take into account the dependence of the noise sensitivity level on background brightness.

Another, more accurate, heuristic mask (M_{bp}) has been built based on the assumption that the watermark will be better hidden in those areas of the image where the watermarked frequencies are present. The watermark building procedure is sketched in Figure 4. The band-pass filter is tuned for extracting just the range of frequencies where the watermark is embedded in; the absolute value of the band-pass filtered image is first thresholded to further highlight these image components; a dilation morphologic filter is then applied to it to fill the holes. In order to enhance the edges a Sobel filter followed by a threshold operation are performed. Furthermore the darker and brighter areas of the original image are highlighted by gray-level thresholding (the following median filters are useful to regularise the thresholding outputs). The four contributions are then added together, and, finally, an offset value is added to the resulting sum in order to allow at least a minimum amount of watermark to be inserted everywhere (also in uniform areas of mean brightness a small level of noise can be tolerated). The mask shown in Figure 3 is thus obtained.

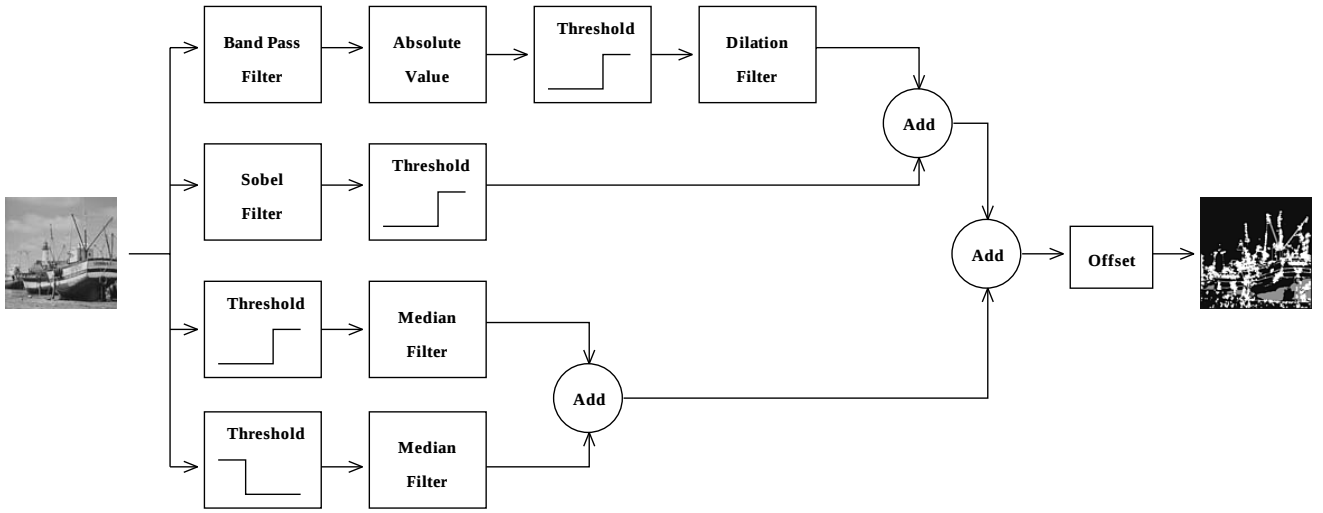


Figure 4: Scheme of the procedure to build the map based on the band-pass filtering of the original image.

5 Experimental results

Experiments have been carried out on many standard images, in the following the results related to *Boat* are reported.

First of all the maximum value of the $\bar{\alpha}$ parameter (see eq. (4)) that can be tolerated without reducing watermark invisibility, has to be considered. In Table 1 such maximum values are reported corresponding to each masking image. It is apparent that the use of a mask (in particular mask M_{bp}) allows to embed a higher energy mark.

Table 1: Maximum values of $\bar{\alpha}$ which can be used with each mask without compromising watermark invisibility.

Masks	$\bar{\alpha}$
No	0.12
M_s	0.13
M_σ	0.13
M_{bp}	0.16

For comparing the performances of the different masks the watermarked images were attacked. The results obtained with two kind of attacks are reported in Table 2. First, images are JPEG compressed at decreasing level of quality until the watermark is still detectable (in Table 2 the minimum achievable percentage quality factor is reported). In the second kind of attack, the image are circularly cropped with a decreasing diameter until the watermark is still detectable. From these experiments appeared that, in

general, the use of the masking images enhances the robustness of the algorithm. In particular it is evident that masks based on theoretically accurate models of the HVS do not offer better results than those based on heuristic considerations. A reason for this can be found in the fact that theoretical models, as that proposed in [9], model well only the behaviour of the eye in the process of human vision, but are very weak in modeling the information integration functions that are performed by the brain. On the contrary masks developed on the basis of heuristic considerations, and experimentally refined, can eventually better simulate the global behaviour (eye+brain) of the HVS.

Experiments were also carried out with multiple attacks. In Table 3 the results obtained with three configurations are reported. In the first case the images are first JPEG compressed and then despeckle filtered; despeckle filter is a powerful tool for adaptively removing disturbs from images without deteriorating their quality. In the second case JPEG compression is followed by circular cropping. In the third case, after JPEG compression, cropping and despeckle filtering, the image is dithered to two colours. The advantages of using the masking procedure are much more evident while performing multiple attacks. In particular it is apparent that watermarking is much more resistant against despeckle if masking is used. Despeckle filter is, in fact, accurately designed not to deteriorate the perceptually important parts of the images (i.e. high activity regions and contours), i.e. the areas where the masking procedure will concentrate more watermarking energy.

In conclusion, the presented approach for masking

Table 2: Results for the experimental comparison of robustness achievable with different masking images: for JPEG the lowest achievable percentage quality factor is given, while for Cropping the smallest tolerable diameter in pixel is reported.

Masks	JPG	Cropping
No	9	330
M_s	7	310
M_σ	5	270
M_{bp}	4	260

Table 3: Results for the experimental comparison of robustness achievable with different masking images in the case of multiple attacks: for JPEG the lowest achievable quality factor is given, while for Cropping the smallest tolerable diameter is reported, and dithering is performed to two colours.

Masks	JPG+Desp.	JPG+Crop.	JPG+Crop.+ Desp.+Dith.
No	47	60 & 350	Not Found
M_s	6	15 & 200	60 & 400
M_σ	6	17 & 200	30 & 400
M_{bp}	4	6 & 200	10 & 300

watermarking signals embedded in the full-frame DCT domain has demonstrated to be effective. The method can be directly extended to watermarking techniques working in the full frame DFT domain.

Future developments will concentrate on the exploitation of colour masking effects to further enhance robustness of the watermarking tool and to allow effective watermark embedding also into chrominance bands.

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